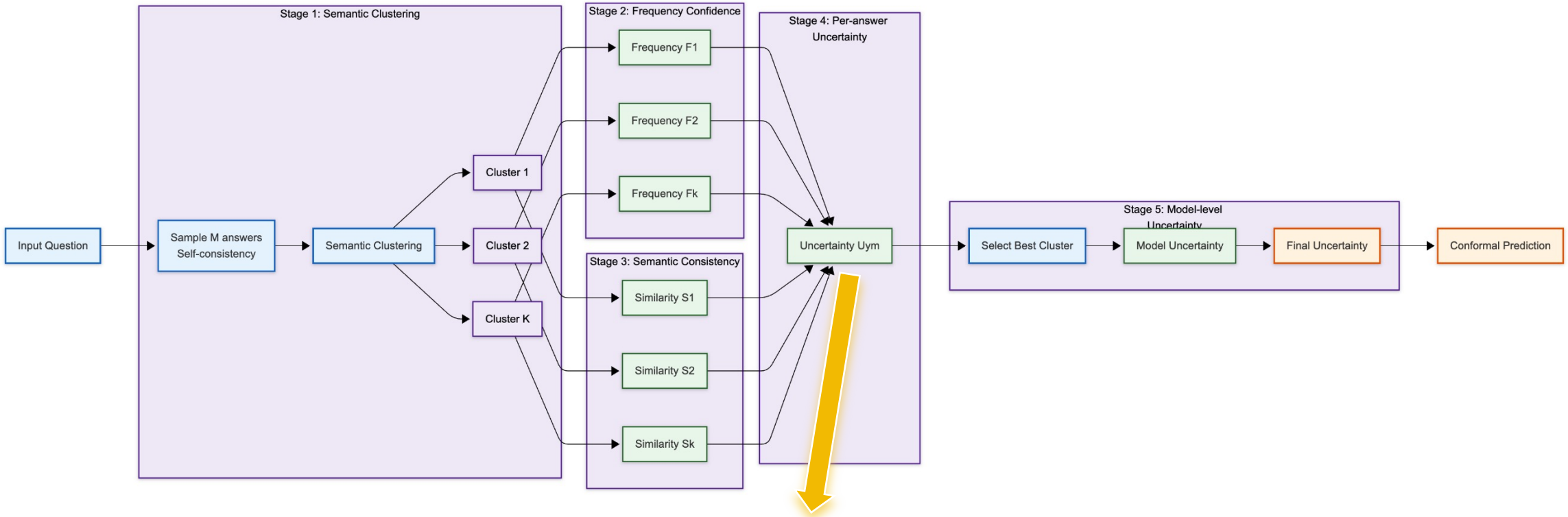
ConU

ConU[108]



ConU measures uncertainty by two complementary components: $\mathcal{F}\left(\hat{y}_k^{(i)}\right) = \frac{V_k}{M}$.

- **Frequency Confidence (F):** How often a semantic meaning appears.
- **Semantic Consistency (S):** How similar each meaning is to others.

$$U\left(\hat{y}_m^{(i)}\right) = 1 - \lambda F\left(\hat{y}_m^{(i)}\right) - (1-\lambda)\frac{1}{K}\sum_{k=1}^{K} S\left(\hat{y}_m^{(i)}, \hat{y}_k^{(i)}\right) F\left(\hat{y}_k^{(i)}\right)$$

cross-encoder model

with $\lambda = 0.5$.

[108] ConU: Conformal Uncertainty in Large Language Models with Correctness Coverage Guarantees, EMNLP 2024

ConU[108]

- **Task Type:** Open-ended generation

- **Introduction:** ConU estimates uncertainty by sampling multiple generations, clustering their semantics, and computing a frequency and semantics based uncertainty score aligned with correctness. It quantifies prediction uncertainty and offers a black-box way to integrate decoding uncertainty into conformal prediction for correctness coverage.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Consensus-based	Decoding	Black-box	Post-hoc	Low	Ranking

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